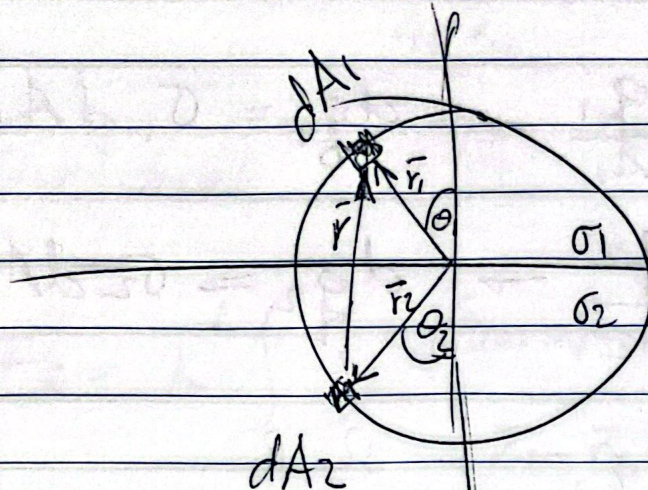


Problema #2: Fuerza entre hemisferios cargados



$$\vec{r}_2 + \vec{r} = \vec{r}_1$$

$$\vec{r} = \vec{r}_1 - \vec{r}_2$$

$$\theta_1: 0 \rightarrow \frac{\pi}{2}$$

$$\theta_2: \frac{\pi}{2} \rightarrow \pi$$

de Coulomb:

$$dF_{12} = \frac{1}{4\pi\epsilon_0} \frac{dq_1 dq_2}{|\vec{r}_1 - \vec{r}_2|^3} (\vec{r}_1 - \vec{r}_2)$$

Por simetría, la fuerza es solo en $\hat{z}$

$$dF_z = \frac{1}{4\pi\epsilon_0} \frac{dq_1 dq_2}{|\vec{r}_1 - \vec{r}_2|^3} (z_1 - z_2)$$

$$z_1 = R \cos \theta_1$$

$$z_2 = R \cos \theta_2$$

Luego

$$\sigma_1 = \frac{q_1}{A_1} \rightarrow dq_1 = \sigma_1 dA_1$$

$$\sigma_2 = \frac{q_2}{A_2} \rightarrow dq_2 = \sigma_2 dA_2$$

Por esféricas

$$dA_1 = R^2 \sin \theta_1 d\theta_1 d\phi_1$$

$$\theta_1: 0 \rightarrow \pi/2$$

$$\phi_1: 0 \rightarrow 2\pi$$

$$dA_2 = R^2 \sin \theta_2 d\theta_2 d\phi_2$$

$$\theta_2: \pi/2 \rightarrow \pi$$

$$\phi_2: 0 \rightarrow 2\pi$$

Entonces

$$dF_z = \frac{1}{4\pi\epsilon_0} \sigma_1 \sigma_2 R^4 \sin \theta_1 d\theta_1 d\phi_1 \sin \theta_2 d\theta_2 d\phi_2 (z_1 - z_2)$$

$$\cos(z_1 - z_2) = R(\cos \theta_1 - \cos \theta_2)$$

llamamos $\frac{1}{4\pi\epsilon_0} \sigma_1 \sigma_2 := K'$

Entonces

$$F_z = K' \int_{S_1} \int_{S_2} \frac{R^5 (\cos\theta_1 - \cos\theta_2)}{|\vec{r}_1 - \vec{r}_2|^3} \sin\theta_1 d\theta_1 d\phi_1 \cdot \sin\theta_2 d\theta_2 d\phi_2$$

$$F_z = K' R^5 \left(\int_0^{2\pi} \int_0^{\pi/2} \int_0^{2\pi} \int_{\pi/2}^{\pi} \frac{(\cos\theta_1 - \cos\theta_2)}{|\vec{r}_1 - \vec{r}_2|^3} \right. \\ \left. * \sin\theta_1 \sin\theta_2 * d\theta_2 d\phi_2 d\theta_1 d\phi_1 \right)$$

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $d\phi_1 \quad d\theta_1 \quad d\phi_2 \quad d\theta_2$

Veamos $|\vec{r}_1 - \vec{r}_2|$

$$\vec{r}_1 = (x_1, y_1, z_1)$$

$$\vec{r}_2 = (x_2, y_2, z_2)$$

En esféricas:

$$\vec{r}_1 = R(\sin\theta_1 \cos\varphi_1; \sin\theta_1 \sin\varphi_1; \cos\theta_1)$$

$$\vec{r}_2 = R(\sin\theta_2 \cos\varphi_2; \sin\theta_2 \sin\varphi_2; \cos\theta_2)$$

$$|\vec{r}_1 - \vec{r}_2|^2 = \vec{r}_1 \cdot \vec{r}_1 + \vec{r}_2 \cdot \vec{r}_2 - 2\vec{r}_1 \cdot \vec{r}_2 \quad (1)$$

$$\text{donde } \vec{r}_1 \cdot \vec{r}_1 = R^2 \quad (2)$$

$$\vec{r}_2 \cdot \vec{r}_2 = R^2 \quad (3)$$

Entonces

$$|\vec{r}_1 - \vec{r}_2|^2 = 2R^2 - 2\vec{r}_1 \cdot \vec{r}_2$$

$$\vec{r}_1 \cdot \vec{r}_2 = R^2 [\sin\theta_1 \cos\varphi_1 \sin\theta_2 \cos\varphi_2 + \sin\theta_1 \sin\varphi_1 \sin\theta_2 \sin\varphi_2 + \cos\theta_1 \cos\theta_2]$$

$$= R^2 [\sin\theta_1 \sin\theta_2 (\underbrace{\cos\varphi_1 \cos\varphi_2 + \sin\varphi_1 \sin\varphi_2}_{\cos(\varphi_1 - \varphi_2)} + \cos\theta_1 \cos\theta_2]$$

$$= R^2 [\sin\theta_1 \sin\theta_2 (\cos(\varphi_1 - \varphi_2)) + \cos\theta_1 \cos\theta_2]$$

(4)

②, ③ y ④ en ①

$$|\vec{r}_1 - \vec{r}_2|^2 = 2R^2 - R^2 \alpha,$$

donde $\alpha = \sin\theta_1 \sin\theta_2 \cos(\varphi_1 - \varphi_2) + \cos\theta_1 \cos\theta_2$

$$|\vec{r}_1 - \vec{r}_2|^2 = \cancel{2R^2} 2R^2 (1 - \alpha)$$

$$|\vec{r}_1 - \vec{r}_2| = \sqrt{2} R \sqrt{1 - \alpha}$$

Subst. en la integral.

$$F_z = K' R^5 \int_0^{2\pi} \int_0^{\pi/2} \int_0^{2\pi} \int_{\pi/2}^{\pi}$$

$$(\cos\theta_1 - \cos\theta_2) \sin\theta_1 \sin\theta_2 d\theta_2 d\varphi_2 d\theta_1 d\varphi_1$$

$$\left[\sqrt{2} R \sqrt{1 - \sin\theta_1 \sin\theta_2 \cos(\varphi_1 - \varphi_2) + \cos\theta_1 \cos\theta_2} \right]^3$$

Resolviendo la integral anterior

con Mathematica:

$$F_z = \frac{\pi r^2}{2\epsilon_0} \sigma_1 \sigma_2$$